Unit 1.1 - Logic

1 Logic Statements

1.1 Definition

A **statement** (also called a proposition) is a sentence that:

- Describes a state of affairs.
- States an alleged fact.
- Communicates a truth value (i.e., it is either true or false).

1.2 Examples of Statements

- "I washed the car."
- "$x = 1$"

1.3 Comparison with Non-Statements

- "What time is it?"
- "Venga aqui!" (Spanish for "Come here!")

1.4 Using Letters to Represent Statements

We often use letters to represent statements, similar to variables in mathematics. For example:

$$p = \text{"I washed the car"}$$

We will then speak of the statement p as being either true or false. We can simplify the claim that p is true by writing the equation $p =$ TRUE instead of writing out he full sentence (same for FALSE).

1.5 Statements that Depend on Variables

The statement "$x \geq 0$" is a statement which depends on a variable. We can write it as $p(x) =$ "$x \geq 0$" (instead of just p) since its truth value will depend on x.

2 Boolean Algebra: Logical Operations and Truth Tables

Boolean algebra was developed by George Boole and forms the foundation of logical operations. Common operations include:

2.1 Negation (NOT)

The negation of a statement p is denoted as $\neg p$ or NOT p.

Example:

$$p = \text{"I went to the store"}$$
$$\neg p = \text{"I did not go to the store"}$$

Truth Table:

A	$\neg A$
True	False
False	True

2.2 Conjunction (AND)

The conjunction of two statements p and q is denoted $p \wedge q$ (read as p **and** q).

Example:

$$p = \text{"I went to the store"}$$
$$q = \text{"I left my house"}$$
$$p \wedge q = \text{"I went to the store AND I left my house"}$$

Truth Table:

p	q	$p \wedge q$
True	True	True
True	False	False
False	True	False
False	False	False

Note that $p \wedge q$ is only TRUE when both inputs are TRUE.

2.3 Disjunction (OR)

The disjunction of two statements p and q is denoted $p \vee q$ (read as p **or** q).

Example:

$$\begin{aligned} p &= \text{"I went to the store"} \\ q &= \text{"I left my house"} \\ p \lor q &= \text{"I went to the store OR I left my house"} \end{aligned}$$

Truth Table:

p	q	$p \lor q$
True	True	True
True	False	True
False	True	True
False	False	False

Note that $p \lor q$ is only FALSE when both inputs are FALSE.

Inclusive vs. Exclusive OR (XOR):

- **Inclusive OR:** $p \lor q$ is true if *at least* one of p or q is true.
- **Exclusive OR:** True if *exactly* one of p or q is true.

2.4 De Morgan's Laws

De Morgan's Laws let you rewrite the negations of conjuntions and disjunctions. They can be proven by computing the truth tables of each side the the equations and seeing that hey are equal.

$$\neg(p \land q) = \neg p \lor \neg q$$
$$\neg(p \lor q) = \neg p \land \neg q$$

2.5 NAND

The NAND operation can be defined as

$$p \text{ NAND } q = \neg(A \land B)$$

All basic logical operations can be rewritten using exclusively NANDs. This is how modern computer chips evaluate logical expressions.

$$\begin{aligned} \neg A &= A \text{ NAND } A \\ A &= \neg(\neg A) = \neg(A \text{ NAND } A) = (A \text{ NAND } A) \text{ NAND } (A \text{ NAND } A) \\ A \land B &= \neg(A \text{ NAND } B) = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B) \\ A \lor B &= \neg((\neg A) \land (\neg B)) = (\neg A) \text{ NAND } (\neg B) = (A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B) \end{aligned}$$

2.6 Implication (If, Then)

Implication, denoted $p \to q$ (read as "if p, then q" or "p implies q"), has an antecedent (p) and a consequent (q).

Example:

$$p = \text{"I went to the store"}$$
$$q = \text{"I left my house"}$$
$$p \to q = \text{"IF I went to the store, THEN I left my house"}$$

Truth Table:

p	q	$p \to q$
True	True	True
True	False	False
False	True	True
False	False	True

Note that $p \to q$ is only FALSE when the antecedent is TRUE and the consequent is FALSE (i.e., a lie).

2.7 Bi-implication (If and Only If, IFF)

Bi-implication, denoted $p \leftrightarrow q$ (read as "p if and only if q"), means that p and q imply each other, i.e., $(p \to q) \wedge (q \to p)$. If either are true/false, then both are true/false.

Truth Table:

p	q	$p \leftrightarrow q$
True	True	True
True	False	False
False	True	False
False	False	True

Note that $p \leftrightarrow q$ is only TRUE when p and q match.

3 Quantifiers

3.1 Universal Quantifier

The universal quantifier, denoted by $\forall$, is used to state that a statament is true of all elements of a certain set.

Example: $\forall x \geq 0, \sqrt{x}$ is defined

This is read as "for all x greater than or equal to 0, the square root of x is defined" (i.e., a real number).

3.2 Existential Quantifier

The existential quantifier, denoted by $\exists$, is used to state that there exists at least one element in a set for which a statement is true.

Example: $\exists x \in R$ such that $x^2 = 2$

This is read as "there exists a real number x such that its square is 2."

If we want to say that there does NOT exist an element with some property, we can strikethrough the symbol: $\nexists$.

Example: $\nexists x \in R$ such that $x^2 = -1$

This is read as "there does not exist a real number x such that its square is -1."

3.3 Negation of Quantifiers

If you want to negate the universal quantifier, you can use an existential quantifier. If we let $p(x)$ be some statement that depends on x (i.e., $\sqrt{x}$ is defined), then we can negate a universal quantifier in the following way

$$\neg(\forall x, p(x)) = \exists x \text{ such that } \neg p(x)$$

In other words, if it isn't the case that $p(x)$ is true for every x, then there must exist at least one value of x for which $p(x)$ is false, i.e., $\neg p(x)$ is true.

Example: $\neg(\forall x \in R, \sqrt{x}$ is defined$) = \exists x \in R$ such that $\sqrt{x}$ is undefined

If you want to negate the existential quantifier, you can use existential quantifier with a strikethrough.

$$\neg(\exists x \text{ such that } p(x)) = \forall x, \neg p(x)$$

In other words, if it isn't the case that there exists some x for which $p(x)$ is true, then $p(x)$ is false (i.e., $\neg p(x)$ is true) for every value of x.

Example: $\neg(\exists x \in R$ such that $\sqrt{x}$ is defined$) = \forall x \in R, \sqrt{x}$ is undefined